SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA1737

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5 **Total Marks: 30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I **Saai Vardhan D / 192525205** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Saai Vardhan D

Annexure A

Resolution Algorithm Implementation

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Resolution Refutation – the method used in every question

Resolution is a single inference rule that works on clauses (disjunctions of literals): from (α ∨ p) and (β ∨ ¬p) infer the resolvent (α ∨ β). It is **sound** (every resolvent is a logical consequence of its parents) and **refutation-complete** (if a set of clauses is unsatisfiable, resolution eventually derives the empty clause □). To prove that a goal Q follows from a knowledge base KB we use **proof by contradiction**:

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1. Represent every statement of KB in propositional logic.
2. Convert KB to CNF:  eliminate α → β  using  ¬α ∨ β ;  move ¬ inward ;
distribute ∨ over ∧.
3. Negate the goal and add it to the clause set:  KB ∧ ¬Q.
4. Repeatedly pick two clauses containing complementary literals (p and ¬p),
   resolve them, and add the resolvent to the set.
5. If the empty clause □ is derived:  KB ∧ ¬Q is unsatisfiable  ⇒  KB ⊨ Q
(goal PROVED).
   If no new clause can be generated and □ was not derived:  goal is NOT
entailed.
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Question 1 – Rain and Wet Ground (6 Marks)

Knowledge Base (KB)

- If it rains, then the ground becomes wet.
- It is raining.

Goal: Prove that the ground is wet.

Step 1 – Represent the statements in propositional logic

Symbol	Proposition
Rain (R)	It is raining
Wet (W)	The ground is wet

Statement	Propositional logic
1. If it rains, then the ground becomes wet	Rain → Wet
2. It is raining	Rain
Goal: the ground is wet	Wet

Step 2 – Convert all statements into CNF

Statement	Implication elimination (α → β ≡ ¬α ∨ β)	CNF clause
Rain → Wet	¬Rain ∨ Wet	(1) ¬Rain ∨ Wet
Rain	(already a literal)	(2) Rain

Step 3 – Negate the goal

Goal: Wet. Negated goal: ¬Wet → (3) ¬Wet.

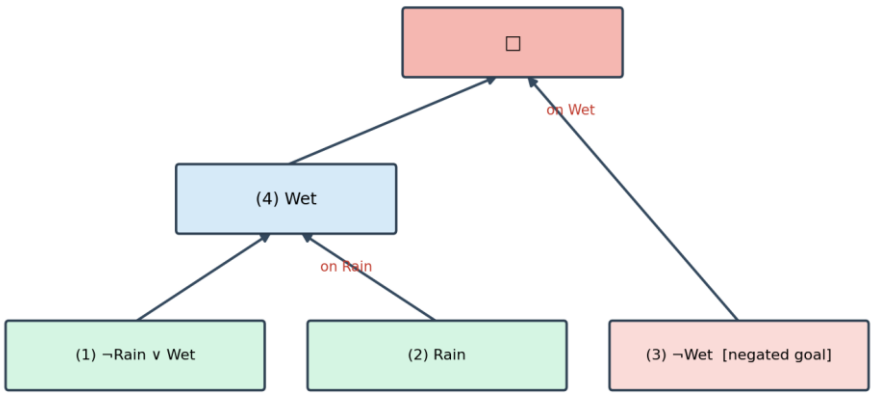
Clause set: (1) ¬Rain ∨ Wet, (2) Rain, (3) ¬Wet.

Step 4 – Apply the Resolution Algorithm step by step

Step	Clauses resolved	Complementary literals	Resolvent
4	(1) ¬Rain ∨ Wet and (2) Rain	Rain / ¬Rain	(4) Wet
5	(4) Wet and (3) ¬Wet	Wet / ¬Wet	(5) □ (empty clause)

Resolution tree

Fig. 1 Resolution tree – Rain and Wet Ground



Step 5 – Conclusion

The empty clause □ is derived, so KB ∧ ¬Wet is a contradiction (unsatisfiable). Hence KB ⊨ Wet – **the goal "the ground is wet" is PROVED.** Semantic check: the only truth assignment that makes both (¬Rain ∨ Wet) and Rain true is Rain = T, Wet = T, so Wet is true in every model of the KB, which agrees with the resolution result.

Question 2 – Student Assignment Submission (6 Marks)

Knowledge Base (KB)

- If a student submits the assignment, then the student receives internal marks.
- Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks.

Step 1 – Represent the statements in propositional logic

Symbol	Proposition
Submit	Rahul submitted the assignment
Marks	Rahul receives internal marks

Statement	Propositional logic
1. If a student submits the assignment, then the student receives internal marks (for Rahul)	Submit → Marks
2. Rahul submitted the assignment	Submit
Goal: Rahul receives internal marks	Marks

Step 2 – Convert all statements into CNF

Statement	Implication elimination ($\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$)	CNF clause
Submit → Marks	¬Submit ∨ Marks	(1) ¬Submit ∨ Marks
Submit	(already a literal)	(2) Submit

Step 3 – Negate the goal

Goal: Marks. Negated goal: ¬Marks → (3) ¬Marks.

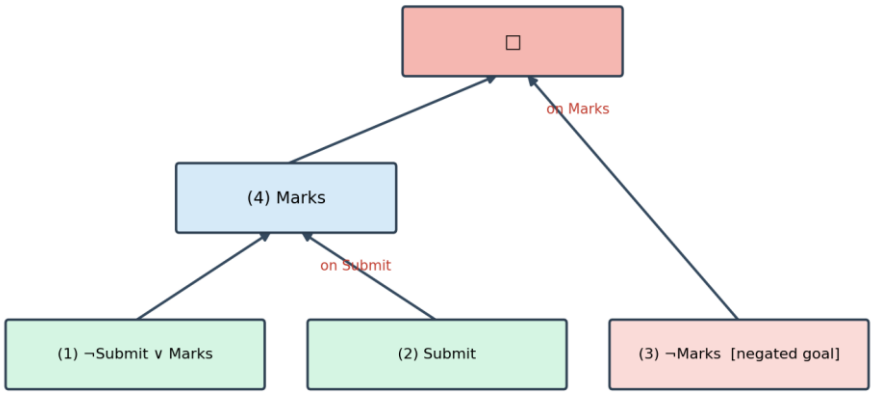
Clause set: (1) ¬Submit ∨ Marks, (2) Submit, (3) ¬Marks.

Step 4 – Apply the Resolution Algorithm step by step

Step	Clauses resolved	Complementary literals	Resolvent
4	(1) ¬Submit ∨ Marks and (2) Submit	Submit / ¬Submit	(4) Marks
5	(4) Marks and (3) ¬Marks	Marks / ¬Marks	(5) □ (empty clause)

Resolution tree

Fig. 2 Resolution tree – Assignment Submission



Step 5 – Conclusion

The empty clause □ is obtained, so the assumption ¬Marks contradicts the knowledge base. Therefore KB ⊨ Marks – **the goal "Rahul receives internal marks" is PROVED.** (The general rule "every student who submits receives marks" can be written in first-order logic as ∀x Submit(x) → Marks(x); instantiating x = Rahul gives the propositional form used here.)

Question 3 – Library Membership (6 Marks)

Knowledge Base (KB)

- If a person is a library member, then the person can borrow books.
- Priya is a library member.

Goal: Prove that Priya can borrow books (and resolve until no new clauses can be generated).

Step 1 – Represent the statements in propositional logic

Symbol	Proposition
Member	Priya is a library member
Borrow	Priya can borrow books

Statement	Propositional logic
1. If a person is a library member, then the person can borrow books (for Priya)	Member → Borrow
2. Priya is a library member	Member
Goal: Priya can borrow books	Borrow

Step 2 – Convert all statements into CNF

Statement	Implication elimination ($\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$)	CNF clause
Member → Borrow	¬Member ∨ Borrow	(1) ¬Member ∨ Borrow
Member	(already a literal)	(2) Member

Step 3 – Negate the goal

Goal: Borrow. Negated goal: ¬Borrow → (3) ¬Borrow.

Clause set: (1) ¬Member ∨ Borrow, (2) Member, (3) ¬Borrow.

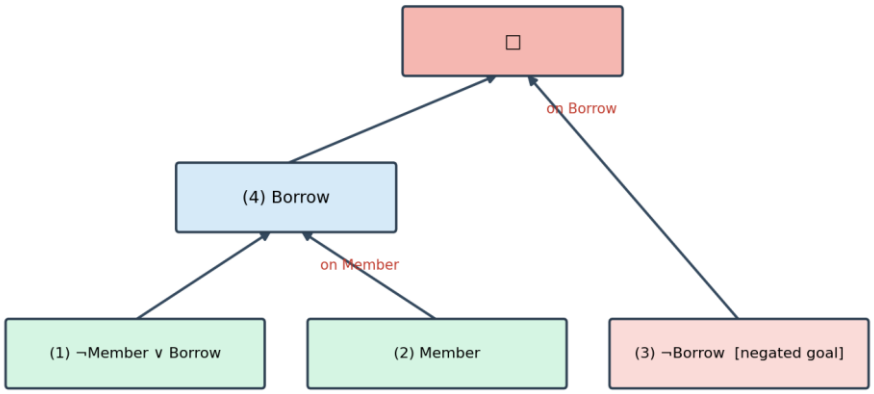
Step 4 – Apply the Resolution Algorithm step by step

Step	Clauses resolved	Complementary literals	Resolvent
4	(1) ¬Member ∨ Borrow and (2) Member	Member / ¬Member	(4) Borrow
5	(1) ¬Member ∨ Borrow and (3) ¬Borrow	Borrow / ¬Borrow	(5) ¬Member
6	(2) Member and (3) ¬Borrow	None (Member vs ¬Borrow)	No resolvent
7	(2) Member and (5) ¬Member	Member / ¬Member	(6) □ (empty clause)
8	(4) Borrow and (3) ¬Borrow	Borrow / ¬Borrow	(7) □ (empty clause)

Resolution until no new clauses can be generated: the complete set of resolvents is {Borrow, ¬Member, □}. As soon as □ appears the refutation is complete (it is reached by two independent routes, steps 7 and 8). Any further pair of clauses either has no complementary literal or only regenerates an existing clause, so the process terminates – **no other new clause is produced**.

Resolution tree

Fig. 3 Resolution tree – Library Membership



Step 5 – Conclusion

The empty clause □ is derived, so KB ∧ ¬Borrow is unsatisfiable. Hence KB ⊨ Borrow – **"Priya can borrow books" is PROVED.** Resolution terminates after a finite number of steps because the set of possible clauses over two symbols is finite.

Question 4 – Placement Eligibility (6 Marks)

Knowledge Base (KB)

- If a student clears the aptitude test, then the student is eligible for placement.
- Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement.

Step 1 – Represent the statements in propositional logic

Symbol	Proposition
Aptitude	Arun cleared the aptitude test
Eligible	Arun is eligible for placement

Statement	Propositional logic
1. If a student clears the aptitude test, then the student is eligible for placement (for Arun)	Aptitude → Eligible
2. Arun cleared the aptitude test	Aptitude
Goal: Arun is eligible for placement	Eligible

Step 2 – Convert all statements into CNF

Statement	Implication elimination (α → β ≡ ¬α ∨ β)	CNF clause
Aptitude → Eligible	¬Aptitude ∨ Eligible	(1) ¬Aptitude ∨ Eligible
Aptitude	(already a literal)	(2) Aptitude

Step 3 – Negate the goal

Goal: Eligible. Negated goal: ¬Eligible → (3) ¬Eligible.

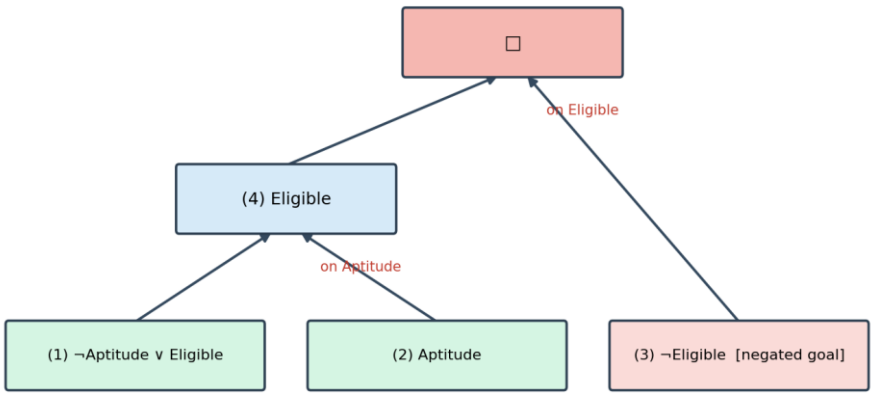
Clause set: (1) ¬Aptitude ∨ Eligible, (2) Aptitude, (3) ¬Eligible.

Step 4 – Apply the Resolution Algorithm step by step

Step	Clauses resolved	Complementary literals	Resolvent
4	(1) ¬Aptitude ∨ Eligible and (2) Aptitude	Aptitude / ¬Aptitude	(4) Eligible
5	(4) Eligible and (3) ¬Eligible	Eligible / ¬Eligible	(5) □ (empty clause)

Resolution tree

Fig. 4 Resolution tree – Placement Eligibility



Step 5 – Conclusion

Reading the tree from the leaves upwards: the KB clauses (1) and (2) give Eligible; Eligible clashes with the negated goal ¬Eligible and yields □. Hence KB ⊨ Eligible – **"Arun is eligible for placement" is PROVED** by resolution refutation.

Question 5 – Access Control System (6 Marks)

Knowledge Base (KB)

- If a user enters the correct password, then the user is authenticated.
- If a user is authenticated, then the user is granted access.
- The user entered the correct password.

Goal: Prove that the user is granted access.

Step 1 – Represent the statements in propositional logic

Symbol	Proposition
Pass	The user entered the correct password
Auth	The user is authenticated
Access	The user is granted access

Statement	Propositional logic
1. If a user enters the correct password, then the user is authenticated	Pass → Auth
2. If a user is authenticated, then the user is granted access	Auth → Access

Statement	Propositional logic
3. The user entered the correct password	Pass
Goal: the user is granted access	Access

Step 2 – Convert all statements into CNF

Statement	Implication elimination ($\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$)	CNF clause
Pass → Auth	¬Pass ∨ Auth	(1) ¬Pass ∨ Auth
Auth → Access	¬Auth ∨ Access	(2) ¬Auth ∨ Access
Pass	(already a literal)	(3) Pass

Step 3 – Negate the goal

Goal: Access. Negated goal: ¬Access → (4) ¬Access.

Clause set: (1) ¬Pass ∨ Auth, (2) ¬Auth ∨ Access, (3) Pass, (4) ¬Access.

Step 4 – Apply the Resolution Algorithm step by step

Step	Clauses resolved	Complementary literals	Resolvent
5	(1) ¬Pass ∨ Auth and (2) ¬Auth ∨ Access	Auth / ¬Auth	(5) ¬Pass ∨ Access (chaining of the two implications)
6	(5) ¬Pass ∨ Access and (3) Pass	Pass / ¬Pass	(6) Access
7	(6) Access and (4) ¬Access	Access / ¬Access	(7) □ (empty clause)

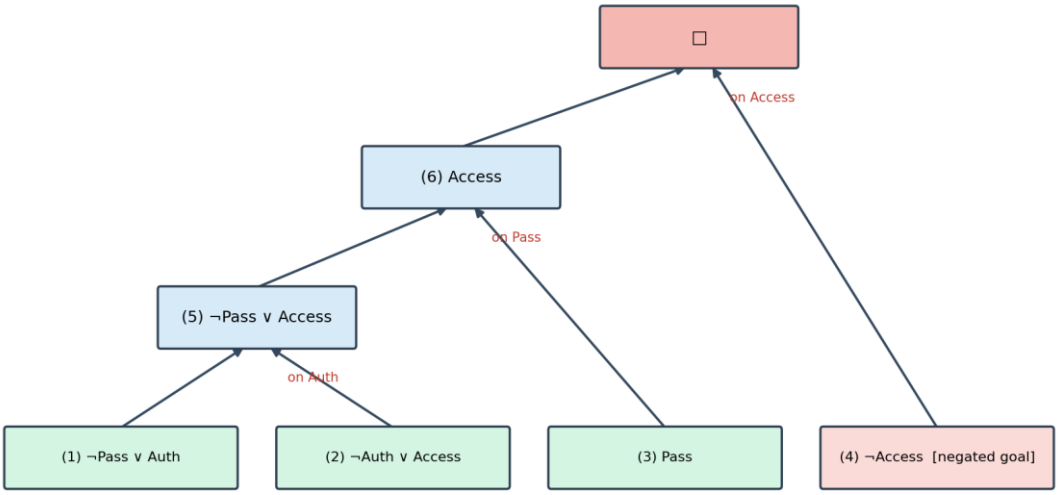
Explanation of each resolution step:

- **Step 5:** the two rules share the symbol Auth (positive in (1), negative in (2)). Resolving removes it and gives ¬Pass ∨ Access, i.e. the derived rule Pass → Access (hypothetical syllogism).
- **Step 6:** the fact Pass cancels ¬Pass in clause (5), leaving the unit clause Access – the user is granted access.
- **Step 7:** Access contradicts the negated goal ¬Access, producing the empty clause □.

An alternative order (using the fact first) gives the same result: (1) and (3) → Auth; Auth and (2) → Access; Access and (4) → □.

Resolution tree

Fig. 5 Resolution tree – Access Control



Step 5 – Conclusion

The empty clause □ is derived, so {Pass → Auth, Auth → Access, Pass} ∧ ¬Access is unsatisfiable. Therefore the KB entails Access – **the user IS granted access (goal PROVED).** Semantic check: in any model where Pass is true, Pass → Auth forces Auth to be true and Auth → Access then forces Access to be true.

Summary of results

Question	Goal	Clauses used	Empty clause derived?	Result
1 – Rain and Wet Ground	Wet	¬Rain ∨ Wet, Rain, ¬Wet	Yes (2 steps)	Proved
2 – Assignment Submission	Marks	¬Submit ∨ Marks, Submit, ¬Marks	Yes (2 steps)	Proved
3 – Library Membership	Borrow	¬Member ∨ Borrow, Member, ¬Borrow	Yes (saturation shown)	Proved
4 – Placement Eligibility	Eligible	¬Aptitude ∨ Eligible, Aptitude, ¬Eligible	Yes (2 steps)	Proved
5 – Access Control	Access	¬Pass ∨ Auth, ¬Auth ∨ Access, Pass, ¬Access	Yes (3 steps)	Proved